

Statement of Work (SOW) Summary for Environmental Testing of Refurbished Flight Spare MOMA WRP

April 11th, 2025

NASA GSFC Code 544

The purpose of this document is to list the work to be performed under the environmental testing contract for the refurbished MOMA flight spare WRP. This document is meant to act as the SOW listing the contractual requirements to be covered under a new NSSC contract.

1. Full Environmental Test Campaign Steps for Refurbished WRP

- a. Bakeout immediately after completing assembly
- b. Comprehensive Performance Tests (CPTs) 1–4 at ambient temperatures per DraMS test procedures and pass/fail criteria. Tests will be performed with the WRP inlet pointed ‘down’ to mimic the final MOMA configuration.
- c. 3-axis sine and random vibration testing to MOMA protoflight levels. Mounting equipment will be provided by GSFC. Test execution shall follow DraMS procedure.
- d. CPTs 1–4 at ambient temperatures. Repeat CPT 4 (verify power draw from < 1 up to 10 Torr back pressure in approximately 2 Torr increments) at MOMA hot and cold allowable flight temperatures (AFTs).
- e. Thermal Cycling to MOMA flight acceptance levels (excluding thermal vacuum cycling). Thermal Cycling test execution shall follow DraMS procedure.
- f. CPTs 1–4 at ambient temperatures.
- g. Final visual cleanliness inspection and packing per DraMS procedures.
- h. Vendor shall make available a full package of items listed under Section 2, Final Deliverables for the environmentally tested WRP at a pre-ship review (PSR) attended by GSFC personnel at Vendor facility prior to acceptance of hardware.
- i. GSFC will accept final WRP assembly by hand at vendor facilities for transport to GSFC.

2. Final Deliverables

- a. CPT Performance Summary report.
- b. Protoflight Vibration and Acceptance Thermal Test Report
- c. Updated Mate and Demate Log
- d. Updated Run Time and Start/Stop Log

NASA shall deliver mounting equipment compatible with the testing house’s vibration tables for vibration testing.

In addition to TRR, and PSR, Vendor shall meet bi-weekly with NASA personnel to discuss test procedure, hardware, and readiness progress until completion of contract.